Mahdy Abdulqadir-Morris

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EDUCATION

University of California, San Diego | **La Jolla, CA**
Bachelors of Science in Mechanical Engineering & Cognitive Science, Machine Learning *June 2027*
Relevant Coursework: Thermodynamics, Fluid Mechanics, Modeling & Data Analysis, Supervised Machine Learning, Neural Networks & Deep Learning, Reinforcement Learning, Calculus, Heat Transfer

TECHNICAL SKILLS

Programming Languages: Java, Python, C , C++, Javascript, MATLAB, SQL
Frameworks and Libraries: Pytorch, Pandas, NumPy
Developer Tools & Practices: Git, Github, Jupyter, Docker
Other tools: Autodesk, Microsoft Office Suite(Excel,Outlook, Word, Powerpoint), Solidworks, Fusion 360, AutoCAD

WORK EXPERIENCE

Northrop Grumman **El Segundo, CA**
Metrology Calibration Intern | January – March 2024

- Calibrated electronic measurement and signal-generation equipment to ensure data accuracy, consistency, and compliance with engineering standards.
- Collected, validated, and documented measurement data, supporting high-quality datasets for downstream engineering and analytical use.
- Applied error analysis, tolerance verification, and statistical reasoning to evaluate instrument performance and measurement reliability.
- Collaborated with cross-functional teams to document (SOPs) and measurement methodologies.
- Assisted engineers and equipment users in designing measurement workflows for complex testing scenarios.
- Utilized computer-based instrument control software and predefined programs, reinforcing experience with automation, system interfaces, and structured data pipelines.

University of California, San Diego **La Jolla, CA**
Transportation Services *August 2022-Current*

- Coordinated setup and breakdown of large-scale campus event signage, ensuring accurate placement based on operational plans and event requirements.
- Utilized Excel spreadsheets and internal systems to log parking activity, track incidents, and support daily operational reporting.
- Supported team operations by following standardized procedures, managing time effectively across multiple locations, and adapting to changing event needs.

PROJECTS

SunGod Robotics | AutoCAD, SolidWorks
Mechanical Robotics Engineer

- Designed, analyzed, and fabricated mechanical components for a competition robot in a multidisciplinary engineering team.
- Modeled parts and assemblies using Fusion 360 and AutoCAD, applying design-for-manufacturing and assembly principles.
- Manufactured and assembled components using TAZ 3D printers and a CO_2 laser cutter, iterating designs based on testing results.
- Developed and tested a two-stage gear-driven lifting mechanism and structural tower components for a robotic arm.
- Validated mechanical performance through functional testing and rapid design iteration to meet competition constraints.

Energy Conservation & Safety Device | Python, Applied ML

- Developed a Python-based intelligent monitoring system to detect appliance misuse and safety risks using multimodal sensor data (motion, light, temperature).
- Engineered feature extraction pipelines from time-series sensor signals to characterize occupancy and activity patterns.
- Implemented lightweight machine learning logic to classify safe vs. unsafe appliance states and trigger automated alerts based on learned behavior rather than fixed thresholds.
- Trained and evaluated simple models (e.g., rule-based classifiers / statistical models) to reduce false positives compared to static thresholding.
- Designed a CAD enclosure to support reliable data collection while protecting system hardware.